

Cassini ISS F Ring Mosaics User Guide

V1.0

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1 Introduction

Saturn's F ring is one of the most dynamic and scientifically intriguing features in the solar system. Discovered in 1979 by the Pioneer 11 spacecraft (Gehrels et al. 1980), the F ring is a narrow, complex, and ever-changing structure located just outside Saturn's main ring system. Its unique morphology, characterized by multiple strands, clumps, and transient features, along with the influence of neighboring moons Prometheus and Pandora, has made it a focal point for studies of planetary ring dynamics, satellite-ring interactions, and the processes governing the evolution of ring systems.

Before 2004, quality images of the F ring were limited to a handful produced by the Voyager 1 and 2 spacecraft during their short fly-bys in 1980 and 1981. This all changed with the arrival of the Cassini Orbiter at Saturn on June 30, 2004. During its subsequent 13 years at Saturn, Cassini used its Imaging Science Subsystem (ISS) (Knowles 2018; Porco et al. 2004) to observe the F ring more than 100,000 times, providing a treasure trove of data that is unlikely to be surpassed for generations.

This *User Guide* provides a detailed description of the organization and contents of the Cassini ISS F Ring Mosaics bundle archived at NASA's Planetary Data System Ring-Moon Systems Node under NASA CDAP grant 80NSSC21K0527: "The recent history of Saturn's dusty rings". The purpose of this archive is to provide a carefully curated and documented set of high-quality images of the F ring taken by the ISS Narrow Angle Camera (NAC) and Wide Angle Camera (WAC) (Porco et al. 2004) using clear filters. Special emphasis is placed on "FMOVIES", sequences of dozens to hundreds of images taken in rapid succession at a single inertial longitude while the F ring orbited through the field of view. Such movies provide a detailed picture of the F ring material at various corotating longitudes as it passes through some true anomaly in the F ring's eccentric orbit.

All included images were calibrated in units of I/F using CISSCAL 4.0 (Knowles 2018) and then reprojected onto a grid with radius relative to the F ring core on the vertical axis and corotating longitude on the horizontal axis (see Section 3.4). Multiple images were stitched together to form more complete mosaics, and the mosaics were then analyzed to remove any background gradient.

The archive includes reprojected images, mosaics, and mosaics that have been adjusted by subtracting a background model, organized by Cassini "observation name" (see Section 3.1). Each reprojected image and mosaic is supplemented by a table containing metadata such as resolution, phase angle, and emission angle for each valid corotating longitude. Reprojected images have an additional file describing the spacecraft pointing used to perform the reprojection. Mosaics have an additional file listing the reprojected images that were used to create the mosaic. A series of browse images in various sizes is provided for each reprojected image and mosaic as well. In total 20,303 reprojected images are provided along with 302 associated mosaics.

Finally, three global index files are provided covering the complete set of reprojected images, mosaics, and background-subtracted mosaics. These index files summarize metadata such as time, corotating longitude coverage, lighting and viewing geometry, and orbit details for each data product, making it easy to search for and analyze the data products *en masse*.

First time users are encouraged to read or skim this entire *User Guide* at least once (and possibly twice) before making use of the archived results.

2 Quick start

The data set described here is archived using the PDS4 standard. While deep knowledge of PDS4 is not required to make use of the data, some familiarity is useful. Specifications, tools, and training are available from the main PDS website at <https://pds.nasa.gov>.¹

This data is archived as a single PDS4 *bundle*, which consists of multiple *collections*. Each collection represents one particular type of data (reprojected images, mosaics, browse images, *etc.*). Within a collection, individual data products are described by a *label* (suffix `.lblx`), which contains substantial information about the data product, including details of any associated binary image (`.img`), graphical image (`.png`), metadata table (`.tab`), or supplemental (`.txt`) files.

Data product files are best read by first processing the associated label file, extracting the needed metadata about the data product, and then using the information in the label to identify the file(s) to read and the details of their format. Special software is available in a variety of programming languages for processing labels and reading the associated data products (Section 5).

Below is a summary of the collections related to reprojected images and the files contained within each.

Collection	Filename(s)	Description
browse_reproj_img (Section 4.2.2)	IMGID_browse_reproj_img.lblx	PDS4 label describing the four types of browse images for reprojected image data.
	IMGID_browse_reproj_img_full.png IMGID_browse_reproj_img_med.png IMGID_browse_reproj_img_small.png IMGID_browse_reproj_img_thumb.png	Human-viewable versions of reprojected images in four sizes in standard PNG format. Images are contrast-stretched and not calibrated for scientific use.
data_reproj_img (Section 4.2.1)	IMGID_reproj_img.lblx	PDS4 label describing the reprojected image and the three associated data files.
	IMGID_reproj_img.img	Binary file containing an $N \times 401$ array of 32-bit floating point values calibrated as I/F. The array represents a grid of corotating longitude in 0.02° increments (x) and radius in increments of 5 km (y). The longitudes are restricted to the range containing valid data.
	IMGID_reproj_img_metadata_params.tab	Fixed-width comma-delimited table containing metadata for each valid corotating longitude in the reproj_img.

¹Refer to the documents in Section 6.1 for further information on the PDS4 concepts of *bundle*, *collection*, and *data product*.

	IMGID_reproj_img_suppl.txt	Text file containing a description of the navigation results for the original Cassini ISS image and the resulting SPICE C matrix.
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Below is a summary of the collections related to mosaics [and background-subtracted mosaics] and the files contained within each.

Collection	Filename(s)	Description
browse_mosaic [original] browse_ mosaic_bkg_sub [background gradient removed] (Section 4.2.4)	OBSID_browse_mosaic[_bkg_sub].lblx	PDS4 label describing the four types of browse images for mosaic data.
	OBSID_browse_mosaic[_bkg_sub]_full.png OBSID_browse_mosaic[_bkg_sub]_med.png OBSID_browse_mosaic[_bkg_sub]_small.png OBSID_browse_mosaic[_bkg_sub]_thumb.png	Human-viewable versions of mosaics in four sizes in standard PNG format. Images are contrast-stretched and not calibrated for scientific use.
data_mosaic [original] data_mosaic_bkg_sub [background gradient removed] (Section 4.2.3)	OBSID_mosaic[_bkg_sub].lblx	PDS4 label describing the mosaic and the three associated data files.
	OBSID_mosaic[_bkg_sub].img	Binary file containing an 18000×401 array of 32-bit floating point values calibrated as I/F. The array represents a grid of corotating longitude in 0.02° increments (x) and radius in increments of 5 km (y).
	OBSID_mosaic[_bkg_sub]_metadata_params.tab	Fixed-width comma-delimited table containing metadata for each valid corotating longitude in the mosaic.

	OBSID_mosaic[_bkg_sub]_metadata_ src_imgs.tab	Fixed-width comma-delimited table containing a map of image numbers from the metadata_params table to the associated reprojected image name.
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Other collections are provided in the bundle, including a document collection containing this User Guide and a miscellaneous bundle containing global index files. All collections and support files are described in the sections below.

The remainder of this User Guide provides detailed guidance on the use of the bundle:

- Section 3 describes the selection and processing of Cassini ISS images and the creation of the mosaics, including the steps:
 - Image selection (Section 3.1)
 - Image calibration (Section 3.2)
 - Image navigation (pointing correction) (Section 3.3)
 - Image reprojection (Section 3.4)
 - Mosaic creation (Section 3.5)
 - Mosaic background gradient modeling and subtraction (Section 3.6)
- Section 4 describes the bundle organization and directory structure, including the details of all files included
- Section 5 discusses techniques for reading label and data product files
- Section 6 contains useful external references
- Section 7 provides an itemized list of all metadata fields and their meaning

3 Image selection and processing

3.1 Image selection

Cassini observations were organized into observing campaigns. Each campaign was given a unique identifier (the observation name), which has the general format:

[INST]_[REV] [TI]_[UNIQUENAME]_[PINST]

where for the observations included in this bundle:

- INST is the abbreviation for the instrument being commanded (usually “ISS”)
- REV is the rev (orbit) number, starting with 000 at Saturn orbit insertion and ending with 293 at the end of mission
- TI is a 2-character target identifier: RA (A Ring), RB (B Ring), RF (F Ring), RI (General Rings), ST (Star)
- UNIQUENAME is a unique descriptor chosen by the observation creator often containing acronyms such as “LP” for low phase, “HR” for high resolution, and “MOV” for movie. Some of the more common names:
 - APOMOS* – Images taken at Cassini apoapse often used to make large mosaics
 - AZSCN* – Azimuthal scan around the rings
 - *OCC – Stellar occultation
 - COMPLIT* – Radial composition scan on the lit side of the rings
 - EGAPMOVMP – Encke gap movie, medium phase
 - FMONITOR – F ring monitor; CIRS scanned the A and B rings while ISS observed the F ring
 - FMOVIE – Stared at a fixed inertial longitude to image F ring material passing through the field of view
 - FMOVIEEQX – F ring movie at Saturn equinox
 - FRSTRCHAN – Prometheus and F ring streamer-channel movie; tracks Prometheus and follows a small section of the F ring around one complete orbit
 - HIRESAFRG – High-resolution A ring outer edge
 - HIRESEFRNG – High-resolution F ring
 - LATPHASE – Latitude and Phase Angle; wide range of sub-spacecraft latitude and solar phase angle
 - PROPRETRG – Re-targeting known “propeller” moons to track their orbits
 - SPK* – Spoke observation; stared at a ring ansa at regular intervals to watch B ring spokes move through the field of view
 - SUBM* – Temperature map taken by CIRS; ISS rider
 - TDIFS* – Temperature map taken by CIRS; ISS rider
 - TMAP* – Temperature map taken by CIRS; ISS rider
 - UR* – UVIS ring stellar occultation; ISS rider
- PINST indicates the name of the prime instrument if other than the ISS (Knowles 2018). If PINST is not “PRIME” then the ISS images were a ride-along on observations designed for the other instrument.

The complete set of ~100,000 images containing the F ring was manually inspected for high-quality images.

In general, an image was included in our data set if:

1. It was a member of a series of images taken at different corotating longitudes that could be stitched together into a mosaic;
2. It was taken with the “clear” filters;
3. It was not corrupted or partially downloaded; and
4. It had a minimum radial resolution for the F ring of at most ~80 km/pixel.

However, due the variation in image attributes and quality, the final determination of which images to include was necessarily subjective. For example:

- Some images were included that do not meet requirement #1. If a single WAC image contained enough of the F ring to subjectively make a good mosaic by itself, it was included. Also, some image sequences were included where the camera followed a single corotating longitude as it traversed different inertial longitudes.
- Some images were included that do not meet requirement #4. These included 42 “SPOKEMOV” sequences taken at relatively low resolution by the WAC. While the reprojected images for these sequences are not of very high quality, the resulting mosaics are nevertheless sufficient for research, such as photometry, that does not require high-resolution analysis.

In total, 20,303 images were chosen for inclusion.

Image sequences were grouped under their original observation name converted to lower case. In some cases, a single observation contained multiple distinct image sequences that would make different mosaics. In these cases, a unique numeric suffix was appended to the observation name.

The selected observations were divided into the classes described in the following sections. Each is indicated in the data archive by a comment in the mosaic PDS4 label and also by a note in the global index file (Section 4.4.1).

3.1.1 *Standard inertial F movies*

A standard “F movie” consists of a series of images taken while Cassini stared at approximately the same inertial longitude. The F ring rotated under the camera for some or all of a complete orbit, providing a partial or complete view of the entire longitudinal extent of the ring. Such observations are considered the default and are thus not marked with a special note.

It is important to realize that it was the F ring that was rotating, not the camera that was instantaneously observing different parts of the ring. As a result, the resulting image sequence and mosaic does not actually provide a view of the F ring around its entire orbit. Instead, it provides repeated views of the F ring centered at (approximately) a single inertial longitude, and thus at a single true anomaly in its eccentric orbit. This can affect the interpretation of morphological features such as streamer-channels, which change shape throughout their orbit (Attree et al. 2014, 2012; Beurle et al. 2010; French et al. 2014a; Murray et al. 2008, 2005).

An example of a standard mosaic is shown in Figure 1.

3.1.2 M#: Observations that are broken into chunks

Although Cassini observing campaigns are indicated by their observation name, sometimes a single campaign produced data that was more appropriately divided into multiple groups. In these cases, we produced multiple mosaics named after the observation name with a numerical suffix. The notes field in the index file (Section 4.4.1) includes “M1”, “M2”, “M3”, or “M4”, indicating how the different chunks are related to each other.

3.1.2.1 M1: Multiple contiguous observations of the same inertial longitude range

In some cases, an observation observed a single inertial longitude for more than a complete orbit of the F ring. In these cases, the observation was broken into multiple chunks and each complete or partial final orbit of the F ring was used to create a separate mosaic. Such observations are noted in the index file with “M1”. The following mosaics have this flag:

- iss_198ri_spokemov005_prime_1/2
- iss_199ri_spokemov002_prime_1/2
- iss_199ri_spokemov003_prime_1/2/3
- iss_199ri_spokemov004_prime_1/2
- iss_199ri_spokemov006_prime_1/2/3
- iss_200ri_spokemov004_prime_1/2/3
- iss_201ri_spokemov001_prime_1/2
- iss_201ri_spokemov009_prime_1/2

3.1.2.2 M2: Pairs of observations taken at inertial longitudes 180° apart

Some observations were performed in pairs, where a particular range of corotating longitudes was observed at one inertial longitude, and then approximately the same range of corotating longitudes was observed at the inertial longitude roughly 180° away. Such pairs provide insight into how particles at the same corotating longitude are actually on slightly different orbits. These observations are divided into two chunks and noted with "M2". The following mosaics have this flag:

- iss_085rf_fmovie003_prime_1/2
- iss_094rf_fmovie001_prime_1/2
- iss_112rf_fmovie002_prime_1/2
- iss_173rf_fmovie001_prime_1/2
- iss_174rf_frstrchan001_prime_1/2
- iss_177rf_fmovie001_prime_1/2
- iss_177rf_frstrchan001_prime_1/2
- iss_183rf_fmovie001_prime_1/2
- iss_193rf_fmovie001_prime_1/2
- iss_194rf_fmovie001_prime_1/2
- iss_198rf_fmovie001_prime_1/2
- iss_201rf_fmovie001_prime_1/2
- iss_202rf_fmovie001_prime_1/2
- iss_203rf_fmovie001_prime_1/2

- iss_208rf_fmovie001_prime_1/2
- iss_208rf_fmovie002_prime_1/2
- iss_209rf_fmovie001_prime_1/2
- iss_212rf_fmovie001_prime_1/2
- iss_232ri_egapmovmp001_prime_1/2
- iss_233rf_fmovie001_prime_1/2
- iss_235rf_fmovie002_prime_1/2
- iss_239rf_fmovie002_prime_1/2
- iss_243rf_fmovie001_prime_1/2
- iss_256rf_fmovie001_prime_1/2
- iss_276ri_hiresafrg001_prime_1/2

An example of such a pair of mosaics is shown in Figure 2.

3.1.2.3 M3: Multiple observations of the same corotating longitude range but different inertial longitude ranges

While standard “F movies” are taken by staring at a single inertial longitude in isolation, some observations were performed by taking multiple “F movies” in a row, each covering roughly the same range of corotating longitudes but at different inertial longitudes. These observations were split apart and are marked with “M3”. The following mosaics have this flag:

- iss_105ri_tmapn45lp001_cirs_1–6
- iss_262rf_fmovie001_prime_01–17
- iss_268rf_fmovie001_prime_2–9

An example is shown in Figure 3.

3.1.2.4 M4: Observations of different corotating and different inertial longitudes

Finally, some observations follow neither a single corotating longitude nor a single inertial longitude. This is often the case for occultations. If both the ingress and egress occultations are imaged, the stellar position in the two has no relation to either the corotating or inertial longitude. There are also some observations that are broken into different segments similar to “M3”, but each chunk observed both different corotating and inertial longitudes. These observations are marked with “M4”. The following mosaics have this flag:

- iss_185ri_rhyaocc001_vims_1/2
- iss_191rf_fmovie001_prime_1/2/3/4/5
- iss_201ri_l2pupocc001_vims_1/2
- iss_206ri_propretrg001_prime_1/2/3
- iss_239ri_hiresafrg001_prime_1/2
- iss_253rf_fmovie001_prime_1/2/3
- iss_268rf_fmovie001_prime_1 (for this mosaic, all of the other chunks are marked with “M3”, but this chunk behaves differently)

3.1.3 R: Follows one corotating longitude range with different inertial longitudes

A small number of observations follow a single corotating longitude, taking one image at each inertial longitude. These are similar to the “M3” observations above, but contain such a small range of corotating longitudes in each image that it is not worth breaking them into separate mosaics. In these cases, a single mosaic is produced, but because the images overlap *the mosaic is not particularly useful*. It is recommended instead to look at the corresponding reprojected images. These observations are marked with “R”. If a single observation is split into chunks, “R” may be paired with an appropriate “M” note. An example is shown in Figure 4.

3.1.4 N: Neither inertial nor corotating

In a few cases, especially those involving wide-field views taken by the WAC instrument, an observation can’t be classified as either inertial or corotating because the camera doesn’t actually follow anything, it just takes a single or short series of images that cover large portions of the F ring. These cases are marked with “N”. An example is shown in Figure 5.

3.1.5 O: Occultation

Several observations were designed to monitor a stellar occultation of the F ring core. These observations continuously observed the position of the star as the F ring rotated. Since each image has the star in it, the mosaic may show the star multiple times. Like the M3 mosaics, these mosaics are *not particularly useful by themselves* and it is recommended to look at the original reprojected images instead. Also, since these are “ride-along” observations where the Cassini Visible and Infrared Mapping Spectrometer (VIMS) or Ultraviolet Imaging Spectrograph (UVIS) instruments were the primary instrument measuring the photon flux, the ISS image sequences do not necessarily show the moment when the star was occulted by the F ring.

The observations ISS_185RI_RHYAOCC001_VIMS and ISS_201RI_L2PUPOCC001_VIMS observed both the ingress and egress and thus are split into two chunks and also marked with “M4”.

The occultation observations are marked with “O”. The following mosaics have this flag.

- iss_172ri_betpegocc001_vims
- iss_172st_urgampeg001_uvis
- iss_180ri_rlyrocc001_vims
- iss_180ri_rcasocc001_vims
- iss_185ri_rhyaocc001_vims_1/2
- iss_194ri_mucepocc001_vims
- iss_196ri_betandocc001_vims
- iss_197ri_whyaocc001_vims
- iss_198ri_rlyrocc001_vims
- iss_201ri_l2pupocc001_vims_1/2
- iss_205ri_l2pupocc002_vims
- iss_206ri_l2pupocc002_vims

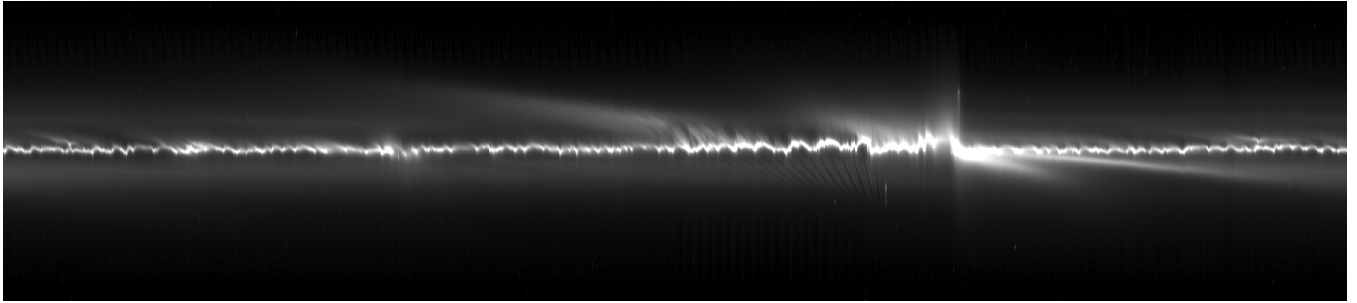


Figure 1: Mosaic of observation `iss_039rf_fmovie001_vims`, a standard “F movie” taken as a rider on a VIMS-prime observation.

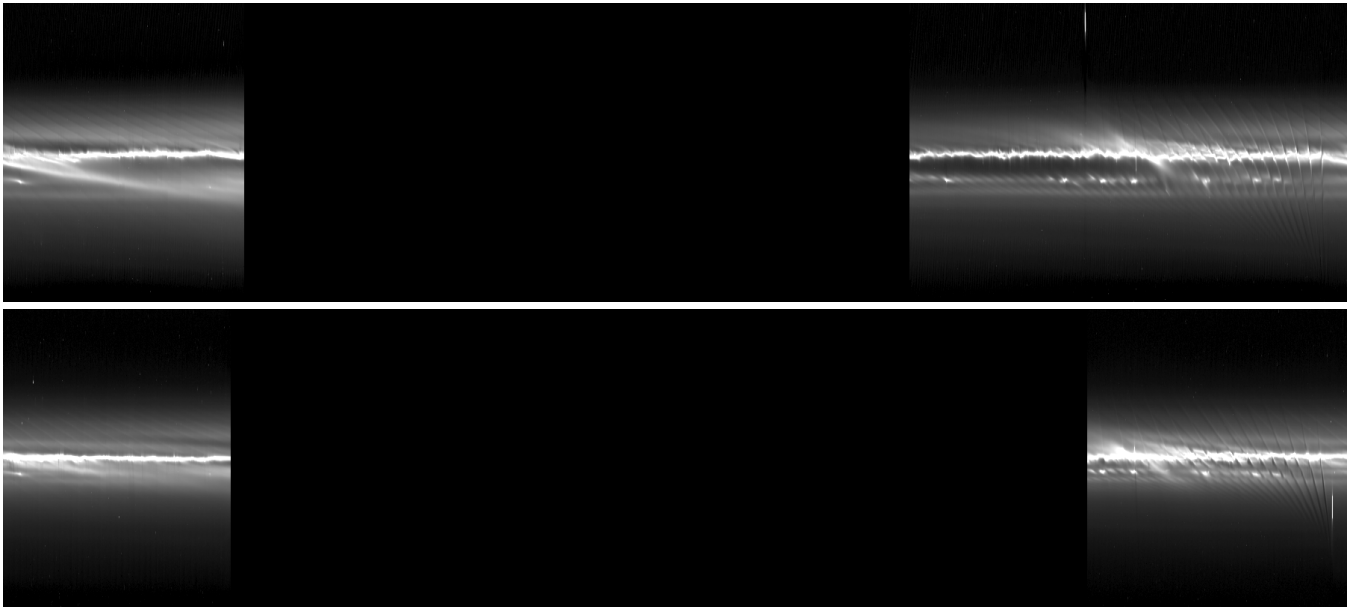


Figure 2: Mosaics of observations `iss_112rf_fmovie002_prime_1` (top) and `iss_112rf_fmovie002_prime_2` (bottom). They cover approximately the same range of corotating longitudes but were taken at inertial longitudes roughly 180° apart. Morphological differences between the two are visible, giving potential insight into the orbits of the particles in these regions. (Flag M2)

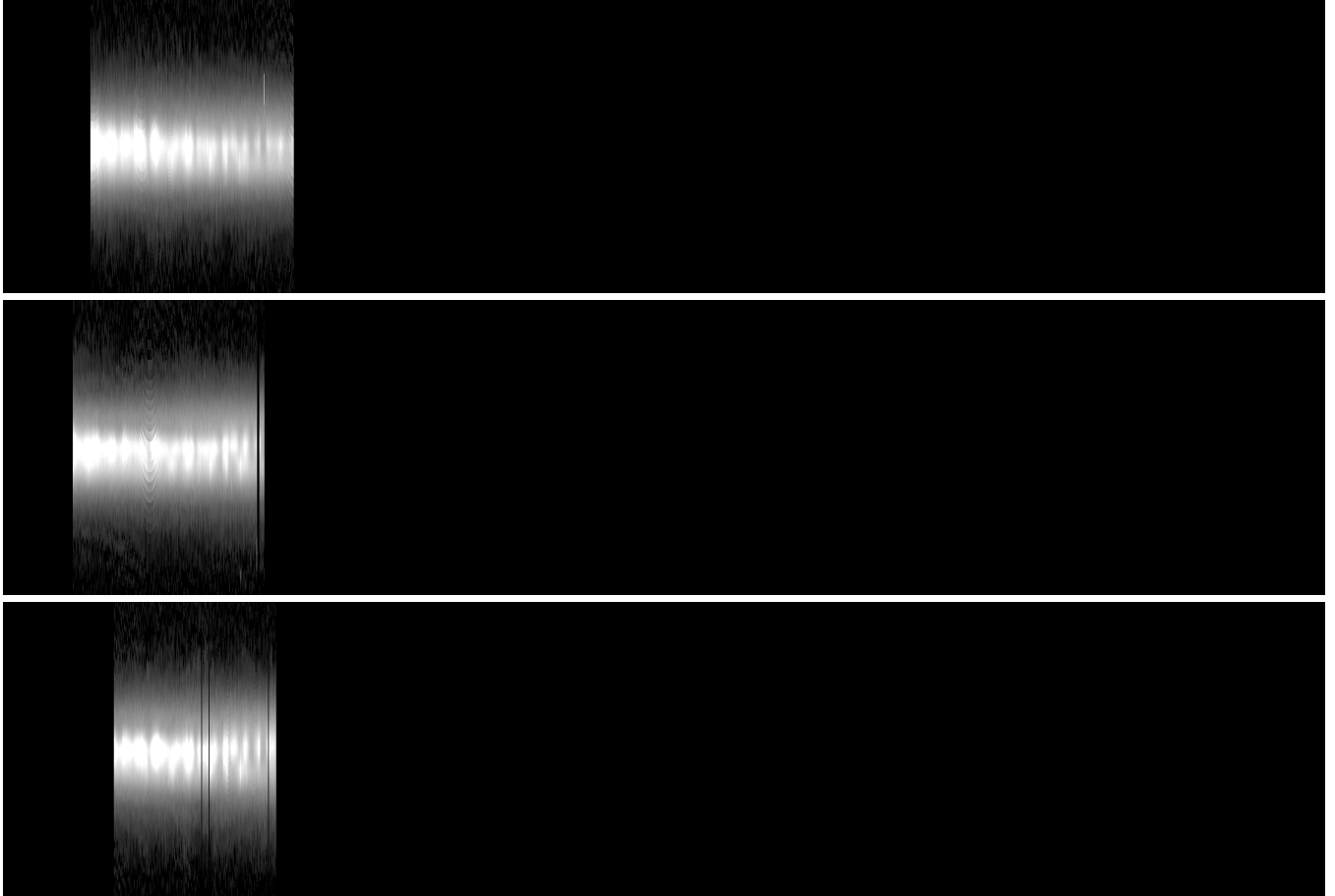


Figure 3: A series of observations covering the same corotating longitudes but taken at progressive inertial longitudes. `iss_105ri_tmapn45lp001_cirs_1` (top) was taken at 78° – 130° ; `iss_105ri_tmapn45lp001_cirs_2` (middle) was taken at 109° – 154° ; `iss_105ri_tmapn45lp001_cirs_3` (bottom) was taken at 136° – 185° . (Flag M3)

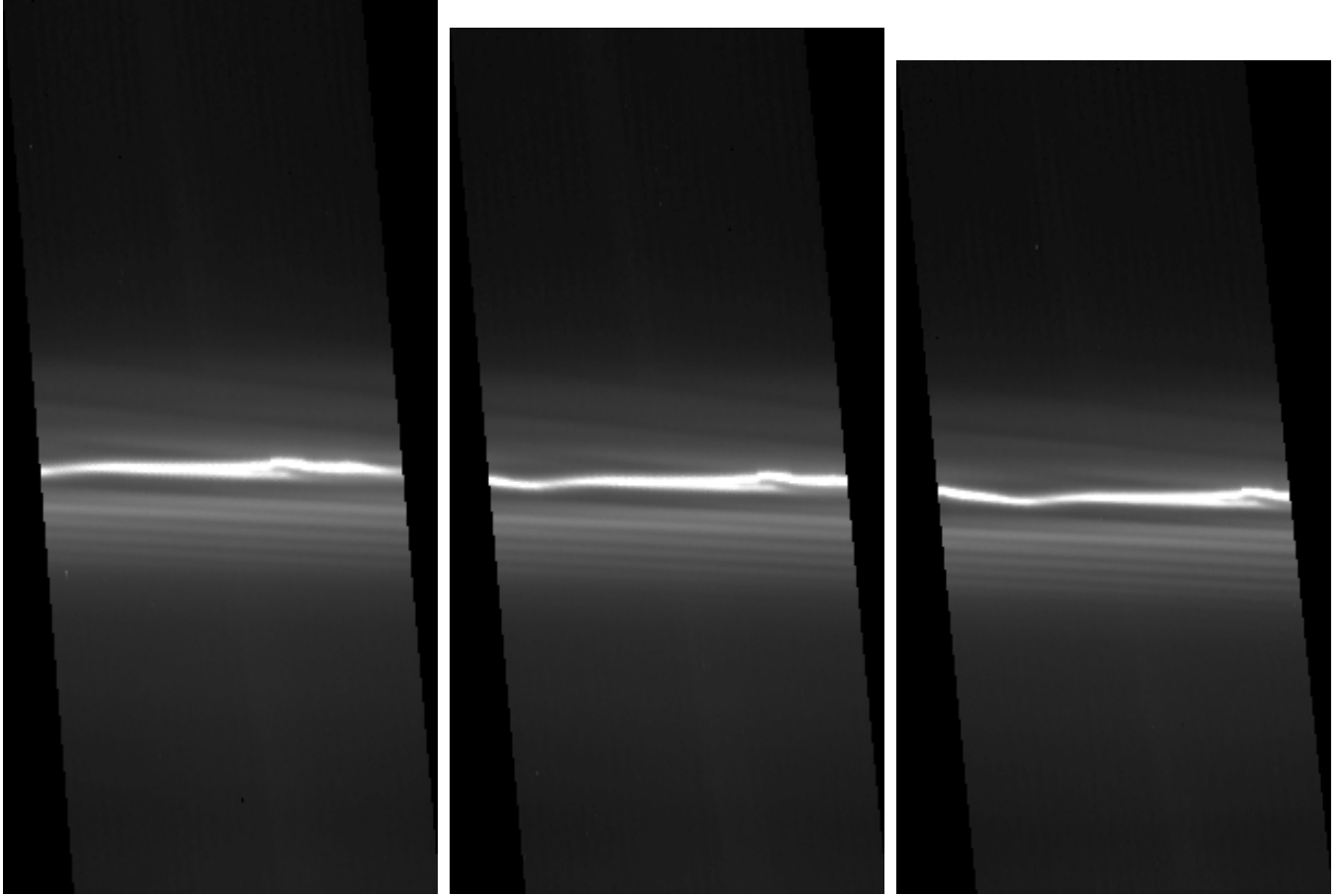


Figure 4: Three reprojected images from mosaic *iss_191rf_fmovie001_prime_1* taken at approximately the same 3° of corotating longitudes. Image #1, 1748745661n (left), was taken at inertial longitudes 6.0° – 9.2° ; image #5, 1748746261n (middle), was taken at inertial longitudes 9.6° – 12.9° ; image #9, 1748746865n (right), was taken at inertial longitudes 13.3° – 16.6° . (Flag R)

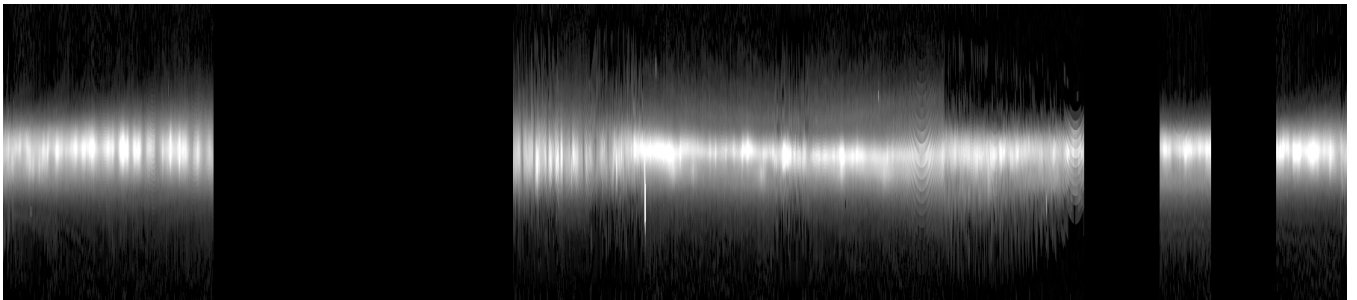


Figure 5: Non-inertial/non-corotating mosaic *iss_098ri_tmapn30lp001_cirs*, which covers most of the full corotating range 0 – 359.98° taken from inertial longitudes 93 – 309° . (Flag N)

3.2 Calibration

Calibrated images were retrieved from the PDS Ring-Moon Systems Node archive². Each image was calibrated by the Node using the CISSCAL 4.0 pipeline (Knowles 2018) into units of I/F³. The calibrated files at the time of writing are being migrated to the PDS4 standard, and those will be archived as collections within the **cassini_iss_cruise** and **cassini_iss_saturn** bundles.⁴

3.3 Navigation

The spacecraft pointing information present in SPICE C kernels is known to have errors on the order of dozens of NAC pixels. Before processing each image, an automatic navigation procedure (French et al. 2014b, 2016) was performed that matched the image with features that were predicted to be present in the image such as stars and the edge of the A ring. Once a mosaic was created from a set of images, the quality of the navigation was evaluated by eye and, in some cases, the navigation of some images was manually adjusted for the best visual result. The (automatically or manually) adjusted C matrix for each image is reported in a supplemental file called **IMGID_reproj_img_suppl.txt** in the **data_reproj_img** collection. See Section 4.2.1.

3.4 Reprojection

Each calibrated image was converted to a reprojected image by mapping the pixels in the source image onto a two-dimensional grid with corotating longitude on the X axis and distance from the nominal F ring core (Δr) on the Y axis.

3.4.1 F ring orbit

Various orbital fits for the F ring core have been developed over time (Albers et al. 2012; Cooper et al. 2013; Cuzzi et al. 2024). We use the orbit fit from (Albers et al. 2012) Table 3 fit #2 throughout:

$$n = 581.964 \text{ }^\circ/\text{day}$$

$$a = 140,221.3 \text{ km}$$

$$e = 0.00235$$

$$\varpi_0 = 24.2^\circ$$

$$\dot{\varpi} = 2.70025 \text{ }^\circ/\text{day}$$

$$\Omega_0 = 15.0^\circ$$

$$\dot{\Omega} = -2.68778 \text{ }^\circ/\text{day}$$

Corotating longitude is based on a reference frame that rotates with the mean motion of the F ring. We use 2007-01-01T00:00:00Z as the epoch when corotating longitude and inertial longitude are equal. To convert from inertial to corotating longitude:

$$\text{corot} = (\text{inertial} - n * (ET - \text{epoch_ET}) / 86400) \% 360$$

and to convert from corotating to inertial longitude:

$$\text{inertial} = (\text{corot} + n * (ET - \text{epoch_ET}) / 86400) \% 360$$

where ET is the ephemeris (SPICE) time of the image in seconds and epoch_ET is the corotating epoch in seconds, equal to 220881665.1839181.

²https://pds-rings.seti.org/holdings/calibrated/COISS_2xxx/

³I/F is a dimensionless value that gives the ratio of the observed intensity to the incoming flux from the light source, assuming a perfect Lambert surface illuminated at normal incidence. See (French et al. 2012) for more details.

⁴https://pds-rings.seti.org/pds4/bundles/cassini_iss

Note that the choice of epoch is arbitrary and when comparing absolute corotating longitudes from different sources it is important to convert to the new reference frame. For example, (Murray et al. 2008) used 2007-01-01T12:00:00Z, (French et al. 2012) used 2004-01-01T12:00:00Z, and (French et al. 2014a) used 2007-01-01T00:00:00Z.

To determine the radius of the F ring core, we use the standard formula:

$$r_{\text{core}} = \frac{a(1 - e^2)}{1 + e \cos(\varpi_0 + \dot{\varpi} \frac{ET}{86400})}$$

3.4.2 Reprojection process

The reprojection grid runs from 0° to 359.98° in steps of 0.02° (~ 49 km) for corotating longitude (18,000 pixels), and -1000 km to $+1000$ km in steps of 5 km for Δr (401 pixels). This gives an aspect ratio of roughly 10:1. For each corotating longitude, the radius is relative to the absolute radius of the core at the associated inertial longitude and time. This means that, while the core of the F ring varies in radius from 139,892 km to 140,551 km throughout its orbit, the core in the reprojected image will always look like an approximately straight line at $\Delta r = 0$, corresponding to $Y = 200$. Note that the odd number of pixels on the Y axis provides an equal number of radii on either side of the core, with the predicted core precisely in the center.

As the majority of reprojected images cover only a few degrees of corotating longitude, archiving files of the complete $18,000 \times 401$ grid would be extremely wasteful. As a result, the reprojected image data products are clipped on the X axis to the minimum and maximum corotating longitudes that contain valid data; see the associated label for information on these values. In the case where the extent of a reprojected image wraps around from 359.98° to 0° , the minimum corotating longitude is larger than the maximum corotating longitude, and the reprojected image is stored in two sections, first from the minimum corotating longitude to 359.98° and then from 0° to the maximum corotating longitude. See Section 5 for examples on how to handle this case in software. In all cases, if there is missing data within the remaining grid (either due to insufficient radial extent available in the source image, or discontinuous corotating longitudes), it is marked with the sentinel value -999 .

During the reprojection process, several other parameters are computed for each pixel:

- Mean phase angle ($^\circ$)
- Mean emission angle ($^\circ$)
- Incidence angle ($^\circ$)
- Mean radial resolution (km/pixel)
- Mean longitudinal resolution ($^\circ$ /pixel)

For each corotating longitude, the mean of each of these parameters for that radial slice is computed and reported in a table `IMGID_reproj_img_metadata_params.tab` (Section 4.2.1) that is associated with each reprojected image. Other parameters describing the orbits of the F ring, Prometheus, and Pandora at the time of the source image are included as well:

- Core Radius (distance from Saturn in km, computed using the formula above)
- Longitude of Ascending Node ($^\circ$)
- Longitude of Pericenter ($^\circ$)
- Min and Max True Anomaly ($^\circ$)

- Corotating Longitude of Prometheus (°) in the same corotating frame used for the F ring
- Radius (distance from Saturn) of Prometheus (km)
- Corotating Longitude of Pandora (°) in the same corotating frame used for the F ring
- Radius (distance from Saturn) of Pandora (km)

Mean, minimum, and maximum values for many of these parameters are included in the label for each reprojected image as well. See Section 7.1 for a complete list of the metadata parameters provided.

3.5 Mosaic creation

Mosaics were created by stitching together one or more reprojected images. The resulting grid for a mosaic is the same $18,000 \times 401$ pixel grid as used for each reprojected image (Section 3.4). For each corotating longitude in the mosaic, the entire radial stripe is taken from a single image. The image is chosen first based on which image has the most valid pixels at that corotating longitude; when there is a tie, the image with the smallest mean radial resolution is used. This guarantees that the resulting mosaic will have the most data and the highest resolution possible, but does not guarantee that the selected reprojected images will change monotonically with corotating longitude. This can result in a slight temporal “blurring” across small ranges of corotating longitude.

Similar to reprojected images, for each corotating longitude several parameters are stored and provided in a table file `OBSID_mosaic_metadata_params.tab` (Section 4.2.3) associated with each mosaic:

- Source reprojected image number
- SPICE ET of source image
- Mean phase angle (°)
- Mean emission angle (°)
- Incidence angle (°)
- Mean radial resolution (km/pixel)
- Mean longitudinal resolution (°/pixel)
- Core Radius (distance from Saturn in km, computed using the formula above)
- Longitude of Pericenter (°)
- True Anomaly (°)
- Corotating Longitude of Prometheus (°) in the same corotating frame used for the F ring
- Radius (distance from Saturn) of Prometheus (km)
- Corotating Longitude of Pandora (°) in the same corotating frame used for the F ring
- Radius (distance from Saturn) of Pandora (km)

Mean, minimum, and maximum values for many of these parameters are included in the label for each mosaic as well. See Section 7.1 for a complete list of the metadata parameters provided.

3.6 Background modeling

The final step in the data production pipeline is modeling and subtracting the background gradient for each mosaic. The background gradient is modeled separately for each corotating longitude using a linear function. Regions normally consisting of the innermost 50 pixels ($\Delta r = -1000$ to -750 km) and the outermost 50 pixels ($\Delta r = +750$ to $+1000$ km) are used to fit a linear model which is then subtracted from the complete 401 pixel radial slice. When creating the model, pixel values that are statistically

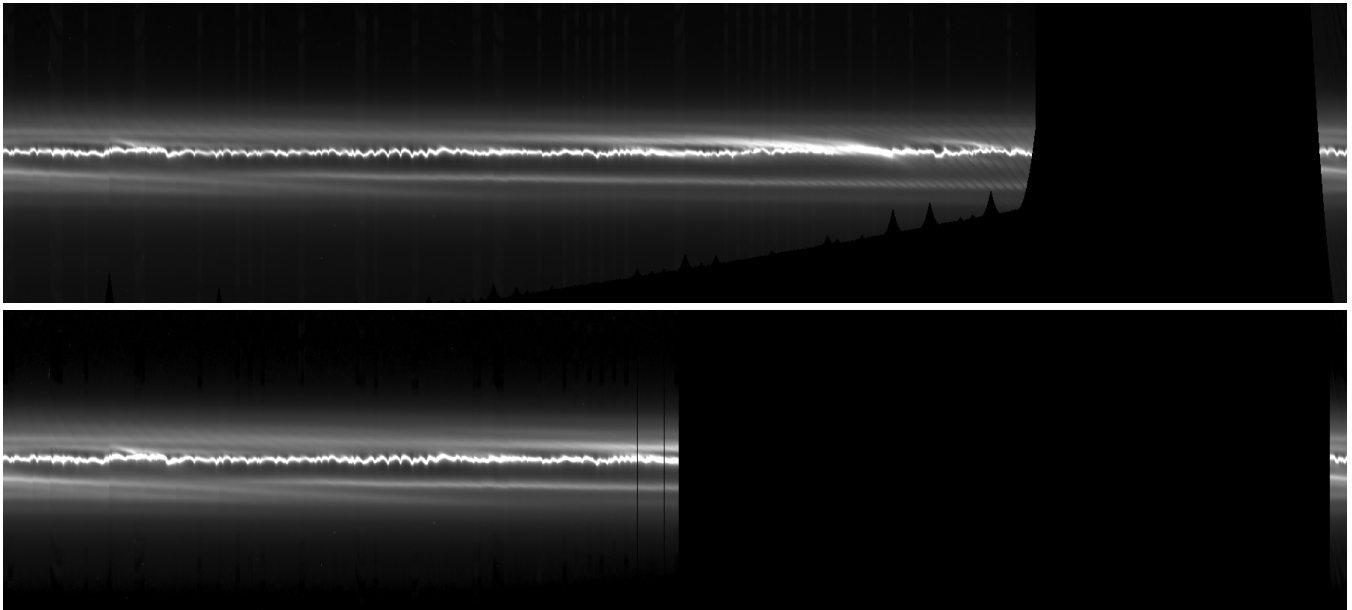


Figure 6: Original mosaic iss_007ri_hpmdfmov001_prime (top) and associated background-subtracted mosaic (bottom). The lack of some data in the original mosaic (to the lower left of the large blank area) prevents the creation of a background model for those longitudes, resulting in a background-subtracted mosaic with substantially fewer valid longitudes.

anomalous compared to nearby longitudes are automatically masked. This removes stars, Prometheus and Pandora, and bad pixels from the background model computation. In some cases, the standard range of 50 pixels on each side is not appropriate. Reasons include large objects such as Prometheus covering the entire 50 pixel range, the reprojected source images not providing data for the entire radial range (which would require the background margins to be increased in size), and very low resolution images where the F ring bleeds over into the background region (which would require background margins to be reduced). Such determinations were made through careful examination of each mosaic and the extent of the background margin was manually adjusted as needed. Whenever possible the new margins were chosen to have minimal effect on computation of equivalent width.⁵ The chosen pixel limits are reported in the label for the background-subtracted mosaic.

In cases where, even after adjusting the pixel limits, there was insufficient radial data to create a background model for some longitudes, those longitudes were marked as invalid and deleted from the background-subtracted mosaic. As a result, the background-subtracted mosaic may have fewer (and sometimes substantially fewer) valid longitudes than the original mosaic. An example is shown in Figure 6.

⁵Equivalent width is a photometric value that integrates the brightness of a radial slice taking into account the resolution of the image.

4 Bundle organization and directory structure

The Cassini ISS F Ring Mosaics *bundle* is organized in a hierarchical directory structure, with each directory containing a PDS4 *collection* of a different type of derived *data product*.⁶ There are collections for reprojected image, mosaics, background-subtracted mosaics, and browse images for each, documentation, miscellaneous (global index files), context, SPICE kernels, and XML schema. The top-level directory structure is:

```
cassini_iss_fring_mosaics_rsfrench2025/
├── bundle.lblx
├── readme.txt
├── browse_mosaic/
├── browse_mosaic_bkg_sub/
├── browse_reproj_img/
├── context/
├── data_mosaic/
├── data_mosaic_bkg_sub/
├── data_reproj_img/
├── document/
├── miscellaneous/
├── spice_kernels/
└── xml_schema/
```

Each directory is described in the sections below. The `context` and `xml_schema` directories are used internally by PDS and will not be discussed further.

For reference, the following types of data products and support files are available:

- **Reprojected images:** These files contain rectangular reprojections of calibrated image files. Like the calibrated files, they are in units of I/F. Associated table files provide information about each longitude. These files are in the `data_reproj_img` directory. See Section 4.2.1.
- **Mosaics:** These files contain mosaics stitched together from multiple reprojected images. Like the reprojected images, they are in units of I/F. Associated table files provide information about each longitude. These files are in the `data_mosaic` directory. See Section 4.2.3.
- **Background-subtracted mosaics:** These files contain the same mosaics as above, but each mosaic has had a linear background model subtracted for each longitude. They are in units of I/F. Associated table files provide information about each longitude. These files are in the `data_mosaic_bkg_sub` directory. See Section 4.2.3.
- **Browse images:** Browse images in four sizes are available for each of the reprojected images, mosaics, and background-subtracted mosaics. These are *not* meant for scientific analysis and are only meant for visual examination of the contents of the data file. The original I/F data has been mapped to integers from 0 to 255 and non-linearly stretched to make dim features easier to see. The details of the stretching process are not provided. These files are in the three `browse_*` directories. See Sections 4.2.2 and 4.2.4.
- **Documentation:** This User Guide is available in PDF format in the `document` directory. See Section 4.3.1.

⁶Refer to the documents in Section 6.1 for further information on the PDS4 concepts of *bundle*, *collection*, and *data product*.

- **Global index files:** These files provide a list of all reprojected images and mosaics in the bundle, along with tabulated metadata for each. They allow for easy programmatic searching on fields such as date and viewing geometry. These files are in the `miscellaneous` directory. See Section 4.4.1
- **SPICE kernels:** A SPICE metakernel specifying the kernels that were used during navigation and data processing is provided in the `spice_kernels` directory. These files are not expected to be needed for most scientific users. See Section 4.5.
- **PDS4 support files:** These files are required by the PDS4 standard and contain information about the contents of the bundle directory. However, they are likely to be less useful to a scientist using the data in the bundle. These files include `bundle.lblx`, `collection_*.csv`, `collection_*.lblx`, and the contents of the `context` and `xml_schema` directories.

4.1 The top-Level directory

The top-level directory contains the following files:

- `bundle.lblx`: The PDS4 bundle label, providing the high-level description of the entire archive.
- `readme.txt`: A text file directing the user to this *User Guide*.

4.2 The *browse* and *data* directories

For each derived data type – reprojected image (Sections 4.2.1 and 4.2.2), mosaic (Sections 4.2.3 and 4.2.4), and background-subtracted mosaic (Sections 4.2.3 and 4.2.4) – there is a `browse` and a `data` directory. These directories have the same internal structure:

```
<dirname>/
├── collection_<dirname>.csv
├── collection_<dirname>.lblx
├── OBSID1/
├── OBSID2/
├── ...
└── OBSIDn/
```

The `collection_<dirname>.csv` file contains a list of all of the data products present for all OBSIDs in this directory, and the accompanying label describes its metadata. The data products themselves are separated by OBSID (including any numeric suffix as discussed in Section 3.1) for ease of retrieval and to keep the number of files in each subdirectory relatively small.

Each derived data type has its own file structure within each OBSID directory.

4.2.1 *data_reproj_img*

This directory contains the individual reprojected images (Section 3.4) organized by OBSID. The directory structure is:

```
data_reproj_img/
├── collection_data_reproj_img.csv
├── collection_data_reproj_img.lblx
├── OBSID1/
│   └── IMGID1_reproj_img.lblx
```

```

├── IMGID1_reproj_img.img
├── IMGID1_reproj_img_metadata_params.tab
├── IMGID1_reproj_img_suppl.txt
└── ...

```

The file `IMGID_reproj_img.lblx` contains all of the details about the reprojected image including the reprojection and viewing geometry:

```

<rings:epoch_reprojection_basis_utc>2007-01-01T00:00:00.000Z</rings:epoch_reprojection_basis_utc>
<rings:reprojection_plane>Equator</rings:reprojection_plane>
<rings:corotating_flag>Y</rings:corotating_flag>
<rings:corotation_rate unit="deg/s">0.006735694444</rings:corotation_rate> <-- 581.964 deg/day -->
<rings:mean_phase_angle unit="deg">36.119</rings:mean_phase_angle>
<rings:minimum_phase_angle unit="deg">35.990</rings:minimum_phase_angle>
<rings:maximum_phase_angle unit="deg">36.182</rings:maximum_phase_angle>
<rings:mean_emission_angle unit="deg">97.421</rings:mean_emission_angle>
<rings:minimum_emission_angle unit="deg">97.200</rings:minimum_emission_angle>
<rings:maximum_emission_angle unit="deg">97.651</rings:maximum_emission_angle>
<rings:mean_incidence_angle unit="deg">88.821387</rings:mean_incidence_angle>
<rings:minimum_incidence_angle unit="deg">88.821387</rings:minimum_incidence_angle>
<rings:maximum_incidence_angle unit="deg">88.821387</rings:maximum_incidence_angle>
<rings:minimum_corotating_ring_longitude unit="deg">286.14</rings:minimum_corotating_ring_longitude>
<rings:maximum_corotating_ring_longitude unit="deg">308.60</rings:maximum_corotating_ring_longitude>
<rings:minimum_ring_radius unit="km">138894.242</rings:minimum_ring_radius>
<rings:maximum_ring_radius unit="km">140934.268</rings:maximum_ring_radius>
<rings:Reprojection_Grid_Parameters>
  <local_identfier>reproj_grid</local_identfier>
  <rings:reprojection_grid_radial_sampling_interval unit="km">
    5.0</rings:reprojection_grid_radial_sampling_interval>
  <rings:reprojection_grid_longitudinal_sampling_interval unit="deg">
    0.02</rings:reprojection_grid_longitudinal_sampling_interval>
  <rings:minimum_radial_resolution unit="km">5.390</rings:minimum_radial_resolution>
  <rings:maximum_radial_resolution unit="km">9.827</rings:maximum_radial_resolution>
  <rings:mean_radial_resolution unit="km">7.001</rings:mean_radial_resolution>
  <rings:minimum_longitudinal_resolution unit="deg">0.016</rings:minimum_longitudinal_resolution>
  <rings:maximum_longitudinal_resolution unit="deg">0.018</rings:maximum_longitudinal_resolution>
  <rings:mean_longitudinal_resolution unit="deg">0.017</rings:mean_longitudinal_resolution>
</rings:Reprojection_Grid_Parameters>

```

and the details of the image file, `IMGID_reproj_img.img`:

```

<File_Area_Observational>
  <File>
    <file_name>1622049830n_reproj_img.img</file_name>
    <creation_date_time>2025-07-25T22:48:10Z</creation_date_time>
    <file_size unit="byte">1050620</file_size>
    <md5_checksum>ad24ba420a8705285effa703a950b61e</md5_checksum>
  </File>
  <-- The calibrated, reprojected single image -->
  <Array_2D_Image>
    <local_identfier>reproj_img</local_identfier>
    <offset unit="byte">0</offset>
    <axes>2</axes>
    <axis_index_order>Last Index Fastest</axis_index_order>
    <Element_Array>
      <data_type>IEEE754LSBSSingle</data_type>
      <unit>I/F</unit>
    </Element_Array>
    <Axis_Array>
      <axis_name>Line</axis_name> <-- vertical (radial) -->
      <elements>401</elements>
      <sequence_number>1</sequence_number>
    </Axis_Array>
  </Axis_Array>

```

```

    <axis_name>Sample</axis_name> <-- horizontal (longitude) -->
    <elements>655</elements>
    <sequence_number>2</sequence_number>
  </Axis_Array>
  <Special_Constants>
    <-- Data was missing or corrupted in the original image (either because it was off
        the edge of the FOV or because of a transmission error resulting in a partial or
        corrupted image) -->
    <missing_constant>-999</missing_constant>
  </Special_Constants>
</Array_2D_Image>
</File_Area_Observational>

```

The image file is a pure binary file with no headers or record separators. Data is stored in LSB IEEE754 single-precision (32-bit) floating point format. The Y axis (Δr) always has size 401 (which covers the radial distance from the core of -1000 km to $+1000$ km in steps of 5 km). The X axis (corotating longitude) has a variable size depending on the number of longitudes that contain valid data (see Section 3.4). Any array elements that do not contain valid data will be marked with the value -999 . See Section 5 for information on how to read these files with software.

An associated metadata file, `IMGID1_reproj_img_metadata_params.tab`, contains information about each *valid* longitude. This table has the general format:

```

rings:corotating_ring_longitude,rings:observed_event_tdb,rings:inertial_ring_longitude,rings:radial_
resolution,rings:longitudinal_resolution,rings:incidence_angle,rings:phase_angle,rings:emission_
_angle,core_radius,longitude_ascending_node,longitude_pericenter,true_anomaly,corotating_longit
ude_prometheus,radius_prometheus,corotating_longitude_pandora,radius_pandora
320.08, 558091398.772, 101.729, 5.232, 0.00010, 63.312, 129.712, 67.484, 140188.617, 293.578,
186.171, 275.558, 177.733, 139483.579, 23.223, 141253.549
320.10, 558091398.772, 101.749, 5.233, 0.00010, 63.312, 129.711, 67.483, 140188.503, 293.578,
186.171, 275.578, 177.733, 139483.579, 23.223, 141253.549
320.12, 558091398.772, 101.769, 5.233, 0.00010, 63.312, 129.710, 67.482, 140188.389, 293.578,
186.171, 275.598, 177.733, 139483.579, 23.223, 141253.549
320.14, 558091398.772, 101.789, 5.233, 0.00010, 63.312, 129.709, 67.481, 140188.274, 293.578,
186.171, 275.618, 177.733, 139483.579, 23.223, 141253.549
320.16, 558091398.772, 101.809, 5.233, 0.00010, 63.312, 129.707, 67.479, 140188.160, 293.578,
186.171, 275.638, 177.733, 139483.579, 23.223, 141253.549[...]

```

where each line refers, in sequence, to an X index in the image array. This table gives the actual corotating longitude that is defined by that index as well as additional useful information such as the inertial longitude, radial resolution, angular (longitudinal) resolution, phase angle, emission angle, and various orbital parameters. See Sections 3.4 and 7 for details about these fields and Section 5 for information on how to read these files with software.

Finally, a supplemental information file, `IMGID1_reproj_img_suppl.txt`, describes the navigation process and the derived pointing for the image (see Section 3.3):

```

Source Data Product ID = 1736795703n_calib
Image Start Time (SCLK) = 1/1736795701.243
Image Start Time (UTC) = 2013-01-13T18:20:56.381Z
Image Mid Time (SCLK) = 1/1736795702.180
Image Mid Time (UTC) = 2013-01-13T18:20:57.131Z
Image Stop Time (SCLK) = 1/1736795703.118
Image Stop Time (UTC) = 2013-01-13T18:20:57.881Z
Trajectory Kernels Query Time = Observation mid time
Stellar Aberration Correction = No
Light Travel Time Correction = No
Navigation Type = Ring and/or Satellite Models
Navigated Boresight RA = 287.227 deg (19h08m54.446s)
Navigated Boresight Dec = 46.7338 deg (+046d44m1.850s)
Navigated Boresight Roll = 22.6952 deg
C-Matrix =

```

0.7979776914	0.5415732280	0.2644428914
-0.5674768892	0.5273925427	0.6323188168
0.2029817339	-0.6546415390	0.7281777744

This pointing information is only provided for archival completeness and is not expected to be used by most scientific users.

4.2.2 *browse_reproj_img*

This directory contains browse images for each reprojected image. The directory structure is:

```

browse_reproj_img/
├── collection_browse_reproj_img.csv
├── collection_browse_reproj_img.lblx
├── OBSID1/
│   ├── IMGID1_browse_reproj_img.lblx
│   ├── IMGID1_browse_reproj_img_full.png
│   ├── IMGID1_browse_reproj_img_med.png
│   ├── IMGID1_browse_reproj_img_small.png
│   ├── IMGID1_browse_reproj_img_thumb.png
│   └── ...
└── ...

```

Four sizes of browse images are provided. “Thumb” images are always 100×100 pixels and are downsampled as necessary from the reprojected image. “Small” images are always 200×200 pixels and are similarly downsampled and padded. “Med” images are downsampled by 10 in the X dimension and are always 400 pixels in the Y dimension, thus having a maximum size of 1800×401. “Full” images are always the same size as the reprojected image (with a minimum of 800 pixels in the X dimension) and guaranteed to be 401 pixels in the Y dimension. All image types are contrast-stretched to use the full available greyscale and are not calibrated for scientific use. Missing data in the reprojected image is indicated by black.

4.2.3 *data_mosaic and data_mosaic_bkg_sub*

These directories have identical structure and contain the original and background-subtracted mosaics, respectively. The directory structures are:

```

data_mosaic/
├── collection_data_mosaic.csv
├── collection_data_mosaic.lblx
├── OBSID1/
│   ├── OBSID1_mosaic.lblx
│   ├── OBSID1_mosaic.img
│   ├── OBSID1_mosaic_metadata_params.tab
│   ├── OBSID1_mosaic_metadata_src_imgs.tab
│   └── ...
└── ...

```

and:

```

data_mosaic_bkg_sub/
├── collection_data_mosaic_bkg_sub.csv
├── collection_data_mosaic_bkg_sub.lblx
├── OBSID1/
│   ├── OBSID1_mosaic_bkg_sub.lblx
│   ├── OBSID1_mosaic_bkg_sub.img
│   ├── OBSID1_mosaic_bkg_sub_metadata_params.tab
│   ├── OBSID1_mosaic_bkg_sub_metadata_src_imgs.tab
│   └── ...
└── ...

```

The format of the `OBSID1_mosaic[_bkg_sub].lblx` and `OBSID1_mosaic[_bkg_sub].img` files is very similar to the corresponding `reproj_img` files described earlier. The only difference is that the mosaic image files are always $18,000 \times 401$ since mosaic images are never clipped to include only valid data (see Sections 3.5 and 4.2.1).

The `OBSID1_mosaic[_bkg_sub]_metadata_params.tab` file is also similar to the corresponding `reproj_img` file, with the exception that it contains one additional column:

```

rings:corotating_ring_longitude,image_index,rings:observed_event_tdb,rings:inertial_ring_longitude,rings:radial_resolution,rings:longitudinal_resolution,rings:incidence_angle,rings:phase_angle,rings:emission_angle,core_radius,longitude_ascending_node,longitude_pericenter,true_anomaly,corotating_longitude_prometheus,radius_prometheus,corotating_longitude_pandora,radius_pandora
308.04, 8, 558093184.000, 81.866, 5.175, 0.00010, 63.312, 129.427, 67.248, 140302.302, 293.523, 186.227, 255.639, 177.819, 139542.008, 23.103, 141339.852
308.06, 8, 558093184.000, 81.886, 5.175, 0.00010, 63.312, 129.426, 67.247, 140302.190, 293.523, 186.227, 255.659, 177.819, 139542.008, 23.103, 141339.852
308.08, 8, 558093184.000, 81.906, 5.175, 0.00010, 63.312, 129.425, 67.245, 140302.078, 293.523, 186.227, 255.679, 177.819, 139542.008, 23.103, 141339.852
308.10, 8, 558093184.000, 81.926, 5.176, 0.00010, 63.312, 129.423, 67.244, 140301.967, 293.523, 186.227, 255.699, 177.819, 139542.008, 23.103, 141339.852
308.12, 8, 558093184.000, 81.946, 5.176, 0.00010, 63.312, 129.422, 67.243, 140301.855, 293.523, 186.227, 255.719, 177.819, 139542.008, 23.103, 141339.852
[...]
```

The new column is “image_index”, which indicates which reprojected image was used to provide data for this particular corotating longitude. The number in the “image_index” column references the file `OBSID1_mosaic[_bkg_sub]_metadata_src_imgs.tab`, which contains a list of the LIDVIDs of the reprojected images that were used to create this mosaic:

```

image_index,LIDVID
0, urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1622022571n_reproj_img::1.0
1, urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1622022704n_reproj_img::1.0
2, urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1622022841n_reproj_img::1.0
[...]
```

While all of these files are in fixed-width format for rapid reading with appropriate software, they can also be read as if they were Comma-Separated Value (CSV) files by standard spreadsheet software. In this case you must specify that the file is delimited by commas.

See Sections 3.4, 3.5, and 7 for details about these fields and Section 5 for information on how to read these files with software.

4.2.4 *browse_mosaic and browse_mosaic_bkg_sub*

These directories contain browse images for the corresponding mosaics. Browse images are provided in four sizes:

- Thumb: 100×100 pixels
- Small: 200×200 pixels
- Medium: 1800×401 pixels
- Full: 18000×401 pixels

Each mosaic is downsampled as necessary to make the browse images. Each browse image is contrast-stretched to use the full available greyscale and is not calibrated for scientific use. Missing data in the mosaic is indicated by black.

4.3 The *document* directory

This directory contains the document collection and has the structure:

```
document/
├── collection_document.csv
├── collection_document.lblx
├── user_guide/
│   ├── display_reproj_img.py
│   ├── f-ring-mosaics-user-guide.lblx
│   ├── f-ring-mosaics-user-guide.pdf
│   ├── find_prometheus_closest_approaches.py
│   ├── plot_ews_df.py
│   └── plot_ews_ma.py
```

4.3.1 *user_guide*

The *user_guide* directory contains the User Guide (this document) and example programs.

4.4 The *miscellaneous* directory – the global index files

This directory contains the global index files and has the structure:

```
miscellaneous/
├── collection_miscellaneous.csv
├── collection_miscellaneous.lblx
├── global_mosaic_bkg_sub_index.lblx
├── global_mosaic_bkg_sub_index.tab
├── global_mosaic_index.lblx
├── global_mosaic_index.tab
├── global_reproj_img_index.lblx
└── global_reproj_img_index.tab
```

4.4.1 *The global index files*

The *miscellaneous* directory contains three global index files. Each index file contains summary information about each reprojected image or mosaic, as appropriate. The provided information is extensive and is designed to be useful to researchers who want to search for images or mosaics meeting certain requirements.

The following notes may be present for each reprojected image and mosaic. If multiple notes are present, they are separated by a semicolon. Each note will also be described in text in the comment section of the label's <Observation_Area>.

- M1: Because Cassini observed the same inertial longitudes for more than one orbit of the F ring during {root_obsid}, we have split the observation into multiple chunks, each corresponding to one complete orbit of the F ring.
- M2: Because observation {root_obsid} consists of two distinct movies consisting of approximately the same corotating longitudes but taken at inertial longitudes roughly 180° apart, we have split the observation into two chunks.
- M3: Because observation {root_obsid} consists of multiple movies consisting of approximately the same corotating longitudes but taken at different inertial longitudes (not 180° apart), we have split the observation into multiple chunks.
- M4: Because Cassini observed multiple distinct inertial and corotating longitudes during {root_obsid}, each making its own movie, we have split the observation into multiple chunks.
- B: This background-subtracted mosaic contains substantially fewer valid longitudes than the original mosaic due to insufficient data being available to create a background model.
- C: Some source images contained corrupted or missing data that could not be repaired during the construction of the mosaic.
- E: Some source images may have been overexposed and the data values clipped; use caution when using this mosaic for photometry.
- O: The sequence of source images used to create this mosaic were designed to observe a stellar occultation of the F ring core. As such, a star is present in each source image and may appear in the mosaic multiple times depending on how the reprojected images were stitched together. In addition, the source images were taken at roughly the same corotating longitudes and thus have significant overlap in the mosaic. To fully explore the occultation, use the reprojected images.
- R: During this time, Cassini followed one corotating longitude for {total_secs} seconds ({total_hours} hours) by observing multiple inertial longitudes covering the (possibly discontinuous) {diff_inertial} degrees from {min_inertial} to {max_inertial}.
- N: During this time, Cassini observed multiple corotating longitudes at multiple inertial longitudes for {total_secs} seconds ({total_hours} hours). The inertial longitudes covered the (possibly discontinuous) {diff_inertial} degrees from {min_inertial} to {max_inertial}.

4.5 The *spice_kernels* directory

This directory contains the SPICE kernel collection, which consists of a single SPICE meta kernel `kernels.ker` listing all kernels used to process the data for this bundle (see Section 6.3). As with the supplemental files described in Section 4.2.1, these files are only provided for archival completeness and are not expected to be used by most scientific users.

5 Reading labels and data product files

5.1 Introduction

The files in this bundle are archived in PDS4 format. PDS4 has the following basic structure:

- A *bundle* consists of one or more *collections*
- A *collection* consists of one or more *data products*
- Each *data product* is defined by PDS4 label (`.lblx`), which gives information about the data and also includes information about the additional associated files that contain the actual data. These data files may be binary or text and may be stored in a variety of formats.

A full discussion of the PDS4 format is beyond the scope of this document and readers are referred to the resources in Section 6.1 for additional reading. However, we will endeavor to provide enough information here to allow reading of these various files for research purposes without comprehensive knowledge of the PDS4 standard.

Two types of software can be used to read PDS4 labels and data files: generic off-the-shelf software that doesn't understand PDS4 concepts, and specific tools or programming libraries that understand PDS4 labels. We will start by discussing the former.

5.2 Using off-the-shelf programs

Label files (`.lblx`), table files (`.tab`), and CSV files (`.csv`) are standard text files that can be read with any text editor. However, while this is sufficient to give you an overview of the contents of the file, each file has a specific format designed to be read and parsed by specific types of programs, and this can be done without the need for PDS4-specific software. Using the correct program in the correct way will make using these files much easier and more productive.

5.2.1 Label files

Label files are written in XML (eXtensible Markup Language), a hierarchical file format much like a nested outline. For the best results, the use of a text editor that understands XML is helpful. Such an editor will likely have two main features:

- The ability to collapse and expand different levels of the hierarchy, so as to hide parts of the label that aren't needed for the task at hand; and
- Syntax coloring, where the open and close tags (`<tagname>data</tagname>`) are color-coded for ease of identification.

As text editors are constantly coming in and out of favor and are platform-specific, we will not recommend specific tools here. However, almost every sophisticated editor or programming IDE (Integrated Development Environment) either natively supports XML or has an XML plugin. In most cases the editor will automatically recognize the file as being in XML format, but in some cases it might be necessary to explicitly invoke the XML mode if the editor relies on the file extension (since `.lblx` is not a normal XML extension).

5.2.2 Table files

Files with the extension `.tab` (including the metadata parameters, source image index, and global indexes) are formatted as *fixed-width comma-delimited tables*. That is, they look like:

```
Value123    ,ValueA  ,ValueB
Value45678  ,ValueCDE,ValueF
```

This format allows software that knows the width of each column to efficiently read the table (see below), but simultaneously permits reading by the wide range of software that supports comma-separated value (CSV) files, such as spreadsheet programs. In this case, when opening the table file, you may find that the program identifies the table as fixed-width. Unfortunately, in this case, it will end up adding the comma to each field. To properly read a table in a spreadsheet program, be sure to specify that the file is “delimited” and that the delimiter is “,”; also specify that you want whitespace stripped if possible.

5.2.3 CSV files

Files with the extension `.csv` (mainly the collection files) are formatted as *variable-width comma-separated value* files. These files can be read the same way as the above table files, except it is more likely programs will correctly identify them as delimited instead of fixed width from the beginning.

5.3 Using PDS4-specific software

A variety of software is available for reading PDS4-format data. Note that over time tools evolve and are replaced, and the URLs or package names where tools can be found change. This section includes tools and their locations at the time of publication. To find new or updated tools, use the resources in Section 6.1

5.3.1 The Small Bodies Node tools

The PDS Small Bodies Node (SBN) has a variety of software useful for reading PDS4-format data. This includes:

- **PDS4 Viewer:** This standalone application is capable of reading PDS4 labels and associated tabular and binary data files and displaying them in various formats.
https://pds-smallbodies.astro.umd.edu/tools/pds4_tools_info/pds4_viewer.html
- **Python PDS4 Tools:** This Python module will read and display PDS4 labels and associated tabular and binary data files. The read data can easily be used in Python programs for further analysis.
https://pdssbn.astro.umd.edu/tools/pds4_tools_info/python_pds4_tools.html
- **ReadPDS:** This IDL module will read and display PDS4 labels and associated tabular and binary data files. The read data can easily be used in IDL programs for further analysis.
https://pdssbn.astro.umd.edu/tools/tools_readPDS.shtml

5.3.2 The Ring-Moon Systems Node tools

The PDS Ring-Moon Systems Node (RMS) provides a Python PDS4 table reader that can also fetch tables from remote locations. It is based on the SBN PDS4 label reader.

<https://rms-pdstable.readthedocs.io/en/latest/>

5.3.3 The Planetary Data Reader

The Planetary Data Reader (PDR) is a Python package that endeavors to be able to read all PDS3 and PDS4 format data with a uniform interface. PDS4 support is based on the SBN PDS4 label reader.

<https://pdr.readthedocs.io/en/latest/>

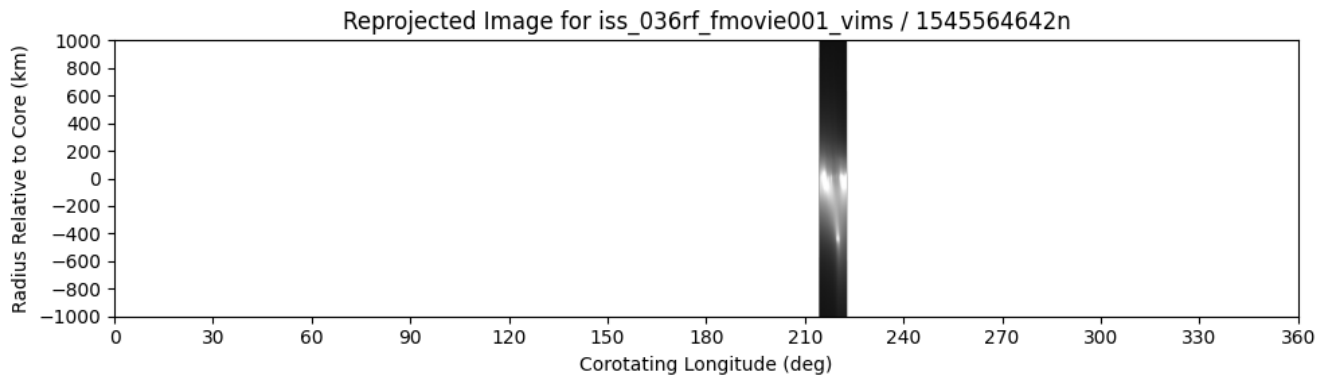
5.3.4 Example programs

The F ring bundle includes example Python programs explicitly designed for reading and manipulating the mosaic and reprojected image data products. Each program is thoroughly commented to aid the user in learning the concepts involved. The programs can be found in the `document/user_guide` directory (Section 4.3.1).

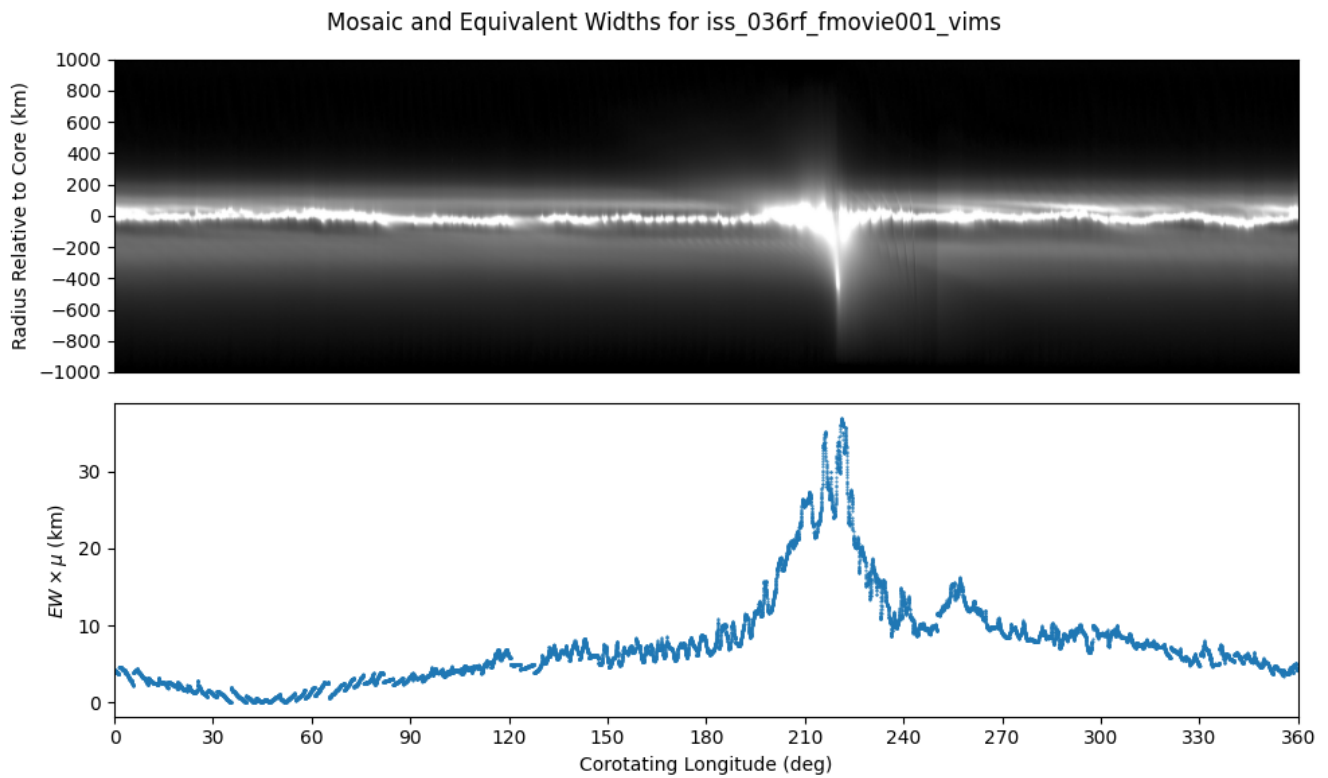
- **mosaic_utils.py:** A support module that is used by all other example programs and that can be directly imported into user-developed programs. It supplies the following functions:
 - **get_element:** Search a label for a particular element name and return its value.
 - **contrast_stretch:** Non-linearly stretch an image based on blackpoint, whitepoint, and gamma values.
 - **read_mosaic_ma** and **read_reproj_img_ma:** Read an F ring mosaic or reprojected image into a numpy masked array with the metadata returned as a structured numpy array.
 - **read_mosaic_ma_df** and **read_reproj_img_ma_df:** Read an F ring mosaic or reprojected image into a numpy masked array with the metadata returned as a pandas DataFrame.
 - **get_mosaic_name_from_lid** and **get_reproj_img_name_from_lid:** Convert a LID to the name of the mosaic or reprojected image.
 - **get_mosaic_name_from_mosaic_label** and **get_reproj_img_name_from_label:** Get the name of the mosaic from the mosaic label or the name of a reprojected image from the reprojected image label.
 - **get_mosaic_name_from_reproj_img_label:** Get the name of the mosaic that incorporates a reprojected image from the reprojected image label.
 - **read_index_np:** Read a global index into a numpy array.
 - **read_index_df:** Read a global index into a pandas DataFrame.
- **display_reproj_img.py:** Display a single reprojected image.
- **plot_ews_ma.py:** Plot a mosaic along with its equivalent width profile using numpy masked arrays for computation.
- **plot_ews_df.py:** Plot a mosaic along with its equivalent width profile using pandas DataFrames for computation (same functionality as the previous program, but the internals are different for illustration).
- **find_prometheus_closest_approaches.py:** Search the global reprojected image index file for images where Prometheus is visible and display the 10 images where Prometheus is closest to the F ring core at that time.

Here are example uses:

```
$ python display_reproj_img.py /path/to/bundle/data_reproj_img/ \
iss_036rf_fmovie001_vims/1545564642n_reproj_img.lblx
```

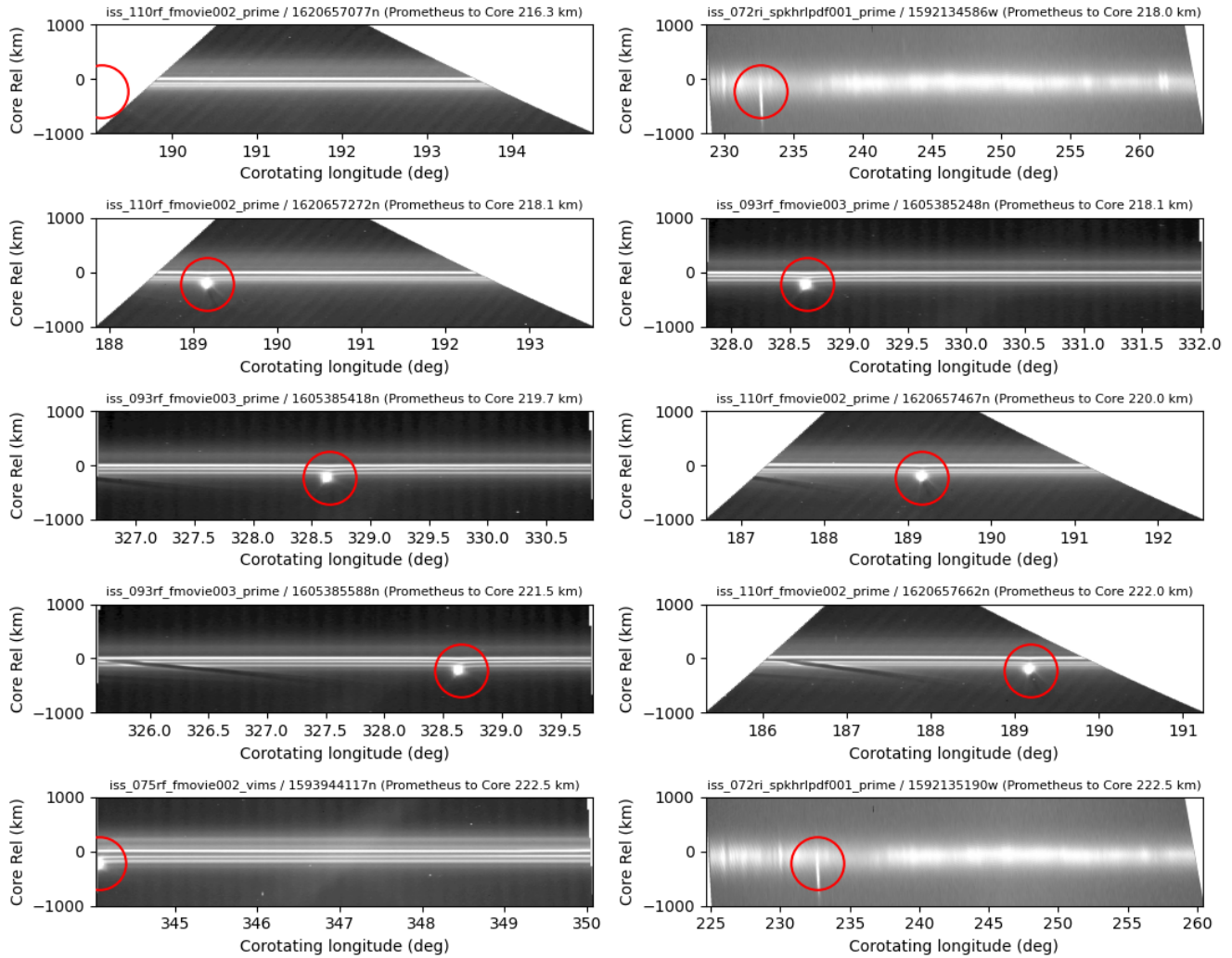


```
$ python plot_ews_ma.py /path/to/bundle/data_mosaic_bkg_sub/ \
iss_036rf_fmovie001_vims/iss_036rf_fmovie001_vims_mosaic_bkg_sub.lblx
```



```
$ python find_prometheus_closest_approaches.py /path/to/bundle
```

Ten closest Prometheus approaches



5.3.5 F ring bundle analysis programs

In addition to the example programs included with the archived bundle, additional software is available to aid researchers in viewing and analyzing the data. As this software evolves rapidly, it is not included with the bundle, but is instead available in this GitHub repo:

<https://github.com/SETI/fring-mosaics-bundle-software>

6 External references and further reading

This section provides a curated list of external references, documentation, and further reading relevant to the Saturn F Ring Cassini ISS Mosaic Bundle.

6.1 PDS4 standards and documentation

- NASA Planetary Data System (PDS) Home: <https://pds.nasa.gov/>
- PDS4 Documents: <https://pds.nasa.gov/datastandards/documents/>
- PDS4 Information Model Specification: <https://pds.nasa.gov/datastandards/documents/im/>
- PDS4 Data Dictionaries: <https://pds.nasa.gov/datastandards/dictionaries/>

6.2 Cassini mission and instrumentation

- Cassini-Huygens Mission Overview: <https://saturn.jpl.nasa.gov/mission/>
- Cassini ISS (Imaging Science Subsystem) Overview: https://pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/Cassini/inst-iss.html
- Cassini ISS Data User Guide (PDS3): <https://doi.org/10.17189/1504135>
- Cassini ISS Narrow Angle Camera (NAC) Details: https://pds-rings.seti.org/pds4/bundles/cassini_iss_saturn/document/iss-na-camera-description.txt
- Cassini ISS Wide Angle Camera (WAC) Details: https://pds-rings.seti.org/pds4/bundles/cassini_iss_saturn/document/iss-wa-camera-description.txt

6.3 SPICE toolkit

- SPICE Toolkit Documentation: <https://naif.jpl.nasa.gov/naif/documentation.html>
- SPICE Tutorials and Reference: <https://naif.jpl.nasa.gov/naif/tutorials.html>

6.4 Additional resources

- PDS Ring-Moon Systems Node: <https://pds-rings.seti.org/>
- PDS Small Bodies Node: <https://pdssbn.astro.umd.edu/>
- PDS Cartography and Imaging Sciences Node: <https://pds-imaging.jpl.nasa.gov/>

6.5 References

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7 Metadata and global index file fields

7.1 Reprojected image and mosaic metadata params

The following fields are tabulated for each valid corotating longitude in each reprojected image (IMG_reproj_img_metadata_params.tab), mosaic (OBSID_mosaic_metadata_params.tab), and background-subtracted mosaic (OBSID_mosaic_bkg_sub_metadata_params.tab).

rings:corotating_ring_longitude	The corotating longitude for which values are being provided
image_index	(Mosaics only) The number of the reprojected image (from OBSID_mosaic_metadata_src_imgs or OBSID_mosaic_bkg_sub_metadata_src_imgs.tab) that provided data for this corotating longitude
rings:observed_event_tdb	The mid-time of the observation that provided data for this corotating longitude (constant for one image)
rings:inertial_ring_longitude	The inertial longitude of the part of the source image that provided data for this corotating longitude
rings:radial_resolution	The radial resolution of the part of the source image that provided data for this corotating longitude
rings:longitudinal_resolution	The angular longitudinal resolution of the part of the source image that provided data for this corotating longitude
rings:incidence_angle	The incidence angle of the ring plane at the time of the source observation (constant for one image)
rings:phase_angle	The phase angle of the part of the source image that provided data for this corotating longitude
rings:emission_angle	The emission angle of the part of the source image that provided data for this corotating longitude
core_radius	The radius of the ideal F ring core from the orbit model at this corotating longitude (constant for one image)
longitude_ascending_node	The longitude of the ascending node (relative to J2000) from the orbit model at the time of the source observation (constant for one image)
longitude_pericenter	The longitude of pericenter (relative to J2000) from the orbit model at the time of the source observation (constant for one image)
true_anomaly	The true anomaly from the orbit model corresponding to this corotating longitude
corotating_longitude_prometheus	The longitude of Prometheus at the time of the source observation in the same frame of reference used for the F ring core (constant for one image)
radius_prometheus	The orbital radius (distance from Saturn center) of Prometheus at the time of the original observation

corotating_longitude_pandora	The longitude of Pandora at the time of the source observation in the same frame of reference used for the F ring core (constant for one image)
radius_pandora	The orbital radius (distance from Saturn center) of Pandora at the time of the original observation

7.2 Reprojected image global index

pds:logical_identifier	The PDS4 logical identifier, <i>e.g.</i> urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1874525875w_reproj_img
cassini:observation_id	The Cassini observation name of the source image, <i>e.g.</i> IOSIC_276RB_COMPLITB3001_SI
file_spec	The path to the reprojected image label file relative to the bundle root, <i>e.g.</i> data_reproj_img/iosic_276rb_complitb4001_si/1874525875w_reproj_img.lblx
pds:start_date_time	The start time of the source image exposure, <i>e.g.</i> 2017-05-26T20:29:10Z
pds:stop_date_time	The stop time of the source image exposure
cassini:spacecraft_clock_start_count	The starting spacecraft clock count of the source image without the partition number (which is always 1/)
cassini:spacecraft_clock_stop_count	The ending spacecraft clock count of the source image without the partition number (which is always 1/)
num_valid_longitudes	The number of discrete longitudes (radial slices) that contain valid data; each radial slice represents 0.02° of corotating longitude
percent_coverage	The percentage of 360° containing valid data; computed from num_valid_longitudes / 18000 * 100
rings:minimum_corotating_ring_longitude	The minimum corotating longitude, representing the left edge of the reprojected image
rings:maximum_corotating_ring_longitude	The maximum corotating longitude, representing the right edge of the reprojected image; note that it's possible for min > max, in which case the corotating range wraps around at 360°
rings:minimum_inertial_ring_longitude	The minimum inertial longitude from the source image used to create the reprojected image

rings:maximum_inertial_ring_longitude	The maximum inertial longitude from the source image used to create the reprojected image; because this is a single exposure, the distance between the min and max corotating longitude and min and max inertial longitude will be identical
rings:mean_phase_angle	The mean (across all valid longitudes) phase angle
rings:minimum_phase_angle	The minimum (across all valid longitudes) phase angle
rings:maximum_phase_angle	The maximum (across all valid longitudes) phase angle
rings:mean_incidence_angle	The ring plane incidence angle at the time of the observation; incidence angle does not vary with longitude so minimum and maximum versions are not supplied
rings:mean_emission_angle	The mean (across all valid longitudes) emission angle; the angle is < 90° on the lit side of the rings
rings:minimum_emission_angle	The minimum (across all valid longitudes) emission angle
rings:maximum_emission_angle	The maximum (across all valid longitudes) emission angle
rings:mean_radial_resolution	The mean (across all valid longitudes) radial resolution in km/pixel; radial resolution is the resolution in the radial direction of the source image used to populate each reprojected image pixel
rings:minimum_radial_resolution	The minimum (across all valid longitudes) radial resolution in km/pixel
rings:maximum_radial_resolution	The maximum (across all valid longitudes) radial resolution in km/pixel
rings:mean_longitudinal_resolution	The mean (across all valid longitudes) angular longitudinal resolution in °/pixel; angular longitudinal resolution is the resolution in the longitude direction (expressed as an angle) of the source image used to populate each reprojected image pixel
rings:minimum_longitudinal_resolution	The minimum (across all valid longitudes) angular longitudinal resolution in °/pixel
rings:maximum_longitudinal_resolution	The maximum (across all valid longitudes) angular longitudinal resolution in °/pixel
rings:minimum_ring_radius	The minimum (across all valid longitudes) ring radius (distance from Saturn center) in the original image used to provide data at any valid longitude
rings:maximum_ring_radius	The maximum (across all valid longitudes) ring radius (distance from Saturn center) in the original image used to provide data at any valid longitude
mean_core_radius	The mean (across all valid longitudes) ring radius (distance from Saturn center) of the ideal F ring core from the orbit model (Section 3.4.1)

minimum_core_radius	The minimum (across all valid longitudes) ring radius (distance from Saturn center) of the F ring core from the orbit model
maximum_core_radius	The maximum (across all valid longitudes) ring radius (distance from Saturn center) of the F ring core from the orbit model
longitude_ascending_node	The longitude of the ascending node of the F ring (relative to J2000) from the orbit model at the time of the original observation (Section 3.4.1)
longitude_pericenter	The longitude of pericenter of the F ring (relative to J2000) from the orbit model at the time of the original observation (Section 3.4.1)
minimum_true_anomaly	The minimum (across all valid longitudes) true anomaly from the orbit model (Section 3.4.1)
maximum_true_anomaly	The maximum (across all valid longitudes) true anomaly from the orbit model (Section 3.4.1)
corotating_longitude_prometheus	The corotating longitude of Prometheus at the time of the original observation in the same reference frame used for the F ring core
radius_prometheus	The orbital radius (distance from Saturn center) of Prometheus at the time of the original observation
corotating_longitude_pandora	The corotating longitude of Pandora at the time of the original observation in the same reference frame used for the F ring core
radius_pandora	The orbital radius (distance from Saturn center) of Pandora at the time of the original observation
pds:creation_date_time	The date and time the reprojected image product was created
nav_quality	The quality of the navigation used for this reprojected image as taken from the mosaic that uses it – this is not a true representation of the navigation quality of this one image, just the overall quality of images in this group and all reprojected images for the same mosaic will have the same indicated quality; values are G=Good, F=Fair, P=Poor (Section 3.3)
notes	Notes about the mosaic that uses this reprojected image (Section 4.4.1)

7.3 Mosaic global index

pds:logical_identifier	The PDS4 logical identifier, <i>e.g.</i> urn:nasa:pds:cassini_iss_fring_mosaics_rsffrench2025:data_mosaic:iosic_276rb_complitb4001_si_mosaic
------------------------	--

cassini:observation_id	The Cassini observation name of the source image, <i>e.g.</i> IOSIC_276RB_COMPLITB3001_SI
file_spec	The path to the mosaic label file relative to the bundle root, <i>e.g.</i> data_mosaic/iosic_276rb_complitb4001_si/iosic_276rb_complitb4001_si_mosaic.lblx
pds:start_date_time	The start time of the earliest source image exposure, <i>e.g.</i> 2017-05-26T20:29:10Z
pds:stop_date_time	The stop time of the latest source image exposure
cassini:spacecraft_clock_start_count	The starting spacecraft clock count of the earliest source image without the partition number (which is always 1/)
cassini:spacecraft_clock_stop_count	The ending spacecraft clock count of the latest source image without the partition number (which is always 1/)
num_valid_longitudes	The number of discrete longitudes (radial slices) that contain valid data; each radial slice represents 0.02° of corotating longitude
percent_coverage	The percentage of 360° containing valid data; computed from num_valid_longitudes / 18000 * 100
rings:minimum_corotating_ring_longitude	The minimum corotating longitude, representing the left edge of the mosaic
rings:maximum_corotating_ring_longitude	The maximum corotating longitude, representing the right edge of the mosaic; note that it's possible for min > max, in which case the corotating range wraps around at 360°
rings:minimum_inertial_ring_longitude	The minimum inertial longitude from the source images used to create the mosaic
rings:maximum_inertial_ring_longitude	The maximum inertial longitude from the source images used to create the mosaic
rings:mean_phase_angle	The mean (across all valid longitudes) phase angle
rings:minimum_phase_angle	The minimum (across all valid longitudes) phase angle
rings:maximum_phase_angle	The maximum (across all valid longitudes) phase angle
rings:mean_incidence_angle	The ring plane incidence angle at the time of the observation; since incidence angle changes slowly over time only a single value is provided
rings:mean_emission_angle	The mean (across all valid longitudes) emission angle; the angle is < 90° on the lit side of the rings
rings:minimum_emission_angle	The minimum (across all valid longitudes) emission angle
rings:maximum_emission_angle	The maximum (across all valid longitudes) emission angle

rings:mean_radial_resolution	The mean (across all valid longitudes) radial resolution in km/pixel; radial resolution is the resolution in the radial direction of the source image used to populate each mosaic pixel
rings:minimum_radial_resolution	The minimum (across all valid longitudes) radial resolution in km/pixel
rings:maximum_radial_resolution	The maximum (across all valid longitudes) radial resolution in km/pixel
rings:mean_longitudinal_resolution	The mean (across all valid longitudes) angular longitudinal resolution in °/pixel; angular longitudinal resolution is the resolution in the longitude direction (expressed as an angle) of the source image used to populate mosaic pixel
rings:minimum_longitudinal_resolution	The minimum (across all valid longitudes) angular longitudinal resolution in °/pixel
rings:maximum_longitudinal_resolution	The maximum (across all valid longitudes) angular longitudinal resolution in °/pixel
rings:minimum_ring_radius	The minimum (across all valid longitudes) ring radius (distance from Saturn center) in the original images used to provide data at any valid longitude
rings:maximum_ring_radius	The maximum (across all valid longitudes) ring radius (distance from Saturn center) in the original images used to provide data at any valid longitude
mean_core_radius	The mean (across all valid longitudes) ring radius (distance from Saturn center) of the ideal F ring core (Section 3.4.1)
minimum_core_radius	The minimum (across all valid longitudes) ring radius (distance from Saturn center) of the F ring core
maximum_core_radius	The maximum (across all valid longitudes) ring radius (distance from Saturn center) of the F ring core
mean_longitude_ascending_node	The mean longitude of the ascending node of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
minimum_longitude_ascending_node	The minimum longitude of the ascending node of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
maximum_longitude_ascending_node	The maximum longitude of the ascending node of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
mean_longitude_pericenter	The mean longitude of pericenter of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
minimum_longitude_pericenter	The minimum longitude of pericenter of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)

maximum_longitude_pericenter	The maximum longitude of pericenter of the F ring (relative to J2000) from the orbit model across all of the source images (Section 3.4.1)
minimum_true_anomaly	The minimum (across all valid longitudes) true anomaly from the orbit model (Section 3.4.1)
maximum_true_anomaly	The maximum (across all valid longitudes) true anomaly from the orbit model (Section 3.4.1)
mean_corotating_longitude_prometheus	The mean corotating longitude of Prometheus in the same reference frame used for the F ring core across all source images
minimum_corotating_longitude_prometheus	The minimum corotating longitude of Prometheus in the same reference frame used for the F ring core across all source images
maximum_corotating_longitude_prometheus	The maximum corotating longitude of Prometheus in the same reference frame used for the F ring core across all source images
mean_radius_prometheus	The mean orbital radius (distance from Saturn center) of Prometheus across all source images
minimum_radius_prometheus	The minimum orbital radius (distance from Saturn center) of Prometheus across all source images
maximum_radius_prometheus	The maximum orbital radius (distance from Saturn center) of Prometheus across all source images
mean_corotating_longitude_pandora	The mean corotating longitude of Pandora in the same reference frame used for the F ring core across all source images
minimum_corotating_longitude_pandora	The minimum corotating longitude of Pandora in the same reference frame used for the F ring core across all source images
maximum_corotating_longitude_pandora	The maximum corotating longitude of Pandora in the same reference frame used for the F ring core across all source images
mean_radius_pandora	The mean orbital radius (distance from Saturn center) of Pandora across all source images
minimum_radius_pandora	The minimum orbital radius (distance from Saturn center) of Pandora across all source images
maximum_radius_pandora	The maximum orbital radius (distance from Saturn center) of Pandora across all source images
pds:creation_date_time	The date and time the mosaic product was created
nav_quality	The overall quality of the navigation across all source images; values are G=Good, F=Fair, P=Poor (Section 3.3)

bkgnd_quality	(Background-subtracted mosaic only) The overall quality of the background model used for this mosaic; values are G=Good, F=Fair, P=Poor (Section 3.6)
notes	Notes about the mosaic (Section 4.4.1)
num_images	The number of reprojected images used to create this mosaic
min_image_name	The name of the earliest source image used in this mosaic
max_image_name	The name of the latest source image used in this mosaic
bkgnd_lower_limit	(Background-subtracted mosaic only) The lower ring delta radius used to delineate the background region; radii between this and core-1000 are used for background modeling.
bkgnd_upper_limit	(Background-subtracted mosaic only) The upper ring delta radius used to delineate the background region; radii between this and core+1000 are used for background modeling.